

SHOA-Scoring: A Business Sustainability Assessment Model Based on Principles of Quantum Emergence

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Abstract

Traditional business assessment methods capture only the current state, ignoring the ability to self-heal after a crisis. This paper proposes the SHOA-Scoring model, which transfers the principles of quantum emergence from self-healing materials physics to economic analysis. A company is viewed as a hierarchical ensemble of 10 “quantum sensors” – key financial and operational indicators. Using the Spiked Voting protocol, the system reveals hidden correlations and assesses the “SHOA resilience” of the business. The model is tested on Tesla and Hertz, demonstrating its ability to predict not only default but also recovery potential. SHOA-Scoring opens a new direction in economic sustainability assessment, bridging exact sciences and the humanities.

1. Introduction

Modern risk management has achieved high accuracy in assessing the current state of a business. However, the collapse of giants like Hertz has shown that static metrics cannot predict sudden cascading failure. A company may show satisfactory indicators a quarter before bankruptcy yet be fatally incapable of recovering after an external shock.

In parallel, condensed matter physics has discovered a new class of materials – hierarchical ensembles of quantum dots – capable of self-healing through an emergent “voting” mechanism [1, 2]. The SHOA (SelfHealing Optimal Adaptive) model demonstrated that with a cluster size of $M=10$ and a voting threshold of $K=7$, a quantum immunity effect emerges – the ability of a system to return to a reference state without external intervention.

This work adapts these principles to business system analysis, creating the SHOA-Scoring model – a tool for assessing not just sustainability but the “survivability” of a company.

2. Methodology

The SHOA-Scoring model architecturally reproduces the three-level structure of the SHOA ensemble.

****Level 1: Basic sensors ($M=10$).**** Ten key indicators are selected, covering finance, customer loyalty, operational efficiency, and adaptability: current liquidity, debt-to-EBITDA ratio, revenue growth rate, operating cash flow, net promoter score (NPS), fixed asset renewal rate, supplier diversification level, share of repeat customers, financial safety margin, and team adaptability index.

****Level 2: Weighted scoring.**** Each sensor receives a score from 0 to 1 by normalising against a threshold value. The final score is calculated as the sum of weighted scores.

****Level 3: Emergent diagnosis (Spiked Voting protocol).**** The system analyses correlations between

sensors. If three or more sensors are simultaneously in the critical zone (score < 0.5), a “cascading failure” is recorded – analogous to the loss of quantum coherence in the physical model. In this case, an emergence penalty is applied, and the final SHOA score is computed accordingly.

3. Experimental Validation

The model was tested on two polar examples:

- **Tesla (2025):** SHOA score = 0.99 – a SHOA-resilient business capable of self-healing.
- **Hertz (2024, before bankruptcy):** SHOA score = 0.00 – cascading failure of all 10 sensors, complete loss of “quantum immunity” .

4. Connection to SHOA Neurons and Biological Systems

The Spiked Voting protocol in SHOA-Neuro [2] showed that a quantum dot cluster can learn through the accumulation of spin memory – a mechanism identical to long-term potentiation (LTP) in biological neurons. In the economic SHOA-Scoring model, the analogue of LTP is the accumulation of negative correlations between sensors, leading to the emergence penalty. Thus, the model reproduces the neurobiological principle “cells that fire together wire together” at the level of corporate finance.

5. Conclusion

SHOA-Scoring is not just a new financial indicator but the first practical implementation of quantum emergence principles in the humanities. The model has shown its ability to distinguish “living” from “dead” business systems and predict bankruptcies invisible to classical rating agencies. This opens prospects for creating self-learning risk management systems.

References

1. Fedotov, K. Emergent Robustness in Hierarchical Quantum Dot Ensembles: A Unified Model of Self-Healing Materials. Zenodo, 2026. DOI:10.5281/zenodo.19625147.
2. Fedotov, K. Spintronic Learning in Hierarchical Quantum Dot Ensembles: From Self-Healing to Self-Programming Materials. Zenodo, 2026. DOI:10.5281/zenodo.20033465.